Networks and Complexity

Solution 13-3

This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at https://github.com/NC-Book/NCB

Ex 13.3: Fibonacci numbers [Lvl. 1]

Compute the first Fibonacci numbers up to F_4 using

$$F_i = \frac{\phi^i - \bar{\phi}^i}{\sqrt{5}}.$$

Convince yourself that this formula produces integer results.

Solution

Since we want to see integer results, we use the exact formula for the golden mean and its little brother

$$\phi = \frac{1+\sqrt{5}}{2} \tag{1}$$

$$\bar{\phi} = \frac{1-\sqrt{5}}{2} \tag{2}$$

We can now compute

$$F_0 = \frac{\phi^0 - \bar{\phi}^0}{\sqrt{5}} = \frac{1-1}{\sqrt{5}} = 0 \tag{3}$$

$$F_1 = \frac{\phi^1 - \bar{\phi}^1}{\sqrt{5}} = \frac{(1+\sqrt{5}) - (1-\sqrt{5})}{2\sqrt{5}} = \frac{2\sqrt{5}}{2\sqrt{5}} = 1 \tag{4}$$

$$F_2 = \frac{\phi^2 - \bar{\phi}^2}{\sqrt{5}} = \frac{(1+2\sqrt{5}+5) - (1-2\sqrt{5}+5)}{4\sqrt{5}} = \frac{4\sqrt{5}}{4\sqrt{5}} = 1 \tag{5}$$

$$F_3 = \frac{\phi^3 - \bar{\phi}^3}{\sqrt{5}} = \frac{(1+3\sqrt{5}+15+5\sqrt{5}) - (1-3\sqrt{5}+15-5\sqrt{5})}{8\sqrt{5}} = \frac{16\sqrt{5}}{8\sqrt{5}} = 2 \tag{6}$$

$$F_4 = \frac{\phi^4 - \bar{\phi}^4}{\sqrt{5}} = \frac{(1+4\sqrt{5}+30+20\sqrt{5}+25) - (1-4\sqrt{5}+30-20\sqrt{5}+25)}{16\sqrt{5}} = \frac{48\sqrt{5}}{16\sqrt{5}} = 3 \tag{7}$$